

Technical Supplement: Activation-Aligned Asynchronous Residual MPC

for Position-Controlled Manipulators with Learned Dynamics

Public evidence bundle for ROBIO 2026

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1 Purpose and evidence boundary

This supplement provides implementation, configuration, and diagnostic detail compressed in the conference manuscript. It introduces no results beyond the frozen or deterministically reanalysed public evidence bundle; its named source files are listed with SHA-256 digests in `PUBLIC.MANIFEST.json`.

All results use MuJoCo position-controlled robot models. ABB IRB 2400 is the primary robot; UR5e is an independently trained and calibrated nominal within-robot replication. The evidence does not establish physical-robot performance, ABB hardware-torque feasibility, hard-real-time scheduling, robust-MPC guarantees, shared-model transfer, or arbitrary-morphology generalization. The post-freeze effort sweep is diagnostic only.

The repository implements an optional budgeted uncertainty supervisor for threaded execution. Model A alone optimizes CEM; compatible GRU replicas assess only the selected short-horizon trajectory, and normalized disagreement can limit residuals or trigger a strict zero-residual fallback to the projected current IK nominal. It was disabled in all frozen experiments, so no uncertainty-detection or safety-performance result is reported.

2 Controller and execution configuration

The controller is a one-layer GRU (hidden size 256; history length 16) that predicts velocity increments from aligned $[q_t, \dot{q}_t, q_t^{\text{ref}}]$ tokens, using one-step increment plus 20-step rollout loss on a disjoint 90/10 split. Frozen controller cost weights/scales are shared; XML models, data distributions, checkpoints, normalizers, references, RobotSpec contracts, and delays are robot-specific. The complete machine-readable contracts are `abb/manifests/control_manifest.json` and `ur5e/manifests/paper.json`.

Table 1: Frozen ABB controller settings. Per-joint vectors and all robot-specific artifacts are in the manifests.

Setting	Value	Setting	Value
Control interval	10 ms	History length	16
Horizon / samples / iterations	20 / 128 / 2	Elite ratio	0.08
Replan interval / delay D	5 / 6 steps	Guard	5 ms
Temporal discount	0.95	Projection	two-stage compiled
$w_q/w_{\dot{q}}/w_r/w_s$	1 / .10 / .20 / .05	$w_{\dot{r}}/w_{\ddot{r}}/w_{\text{first}}$	.05 / .02 / .20
Joint / velocity barrier	10 / 5	Barrier max multiplier	2
Task position/orientation	1 / .25	Task scales	.05 m / .0873 rad
Feedback $K_q/K_{\dot{q}}/\delta r_{\text{max}}$	.3 / .015 / .015 rad	Residual maximum	(0.12,.10,.12,.15,.15,.20) rad

True-history window validation. Held-out model errors use fixed windows beginning at steps 16, 36, $\dots$, 176 of each saved 200-step rollout. Every window therefore has a complete 16-token ground-truth recurrent context before its first predicted successor. Each horizon is evaluated as a separate segmentwise RMSE: $h = 1$ measures the first successor and $h = 20$ pools prediction steps 1:20 within each window. These are not terminal errors. The historical start-of-rollout validation, which padded the missing initial GRU history, is retained in the evidence bundle for audit but is not used in the manuscript table.

Residual packets encode corrections, not reusable absolute commands. At packet age a , the full protocol reconstructs and projects

$$q_t^{\text{cmd}} = \Pi_{v,a,\mathcal{Q}}(\bar{q}_t^{\text{IK}} + r_a + \delta r_t; q_{t-1}^{\text{cmd}}, \dot{q}_{t-1}^{\text{cmd}}).$$

The launch-anchor replay counterfactual instead projects the cached planned absolute command q_a^{plan} . Its large failure is therefore evidence for the coordinate transformation required by an independently evolving nominal, not a competitive controller comparison. The implementation is in `mpc/delay_aware_runner.py`; the protocol definitions are in `mpc/delay_protocol.py`.

3 Artifacts, references, and calibration

The manifests record robot contracts, XML/source hashes, command limits, model, data, and reference hashes, and activation delays. For UR5e, legacy base-run arguments retain pre-resolution workflow values for fingerprint compatibility; separate fields record the resolved robot contract and formal per-case delays. The guard-adjusted 95th-percentile calibration gives ABB $D = 6$ and UR5e $D = 7$. Checkpoint and normalizer SHA-256 digests are recorded in the manifests; artifacts themselves are not committed to avoid distributing large learned model files through Git.

4 Full diagnostics

Table 2: Held-out open-loop model errors from complete-history windows, pooled over prediction steps $1:h$. Errors are $q/\dot{q}$ RMSE in $\text{rad}/\text{rad s}^{-1}$; Div. is the divergent-window rate at $\sigma_u = 0.8$.

Robot, h	q RMSE, $\sigma_u = .5$	$\dot{q}$ RMSE, $\sigma_u = .5$	q RMSE, $\sigma_u = .8$	$\dot{q}$ RMSE, $\sigma_u = .8$	Div., $\sigma_u = .8$
ABB, 1	.000670	.001990	.001069	.005570	0%
ABB, 20	.004852	.034472	.006876	.051398	0%
UR5e, 1	.000132	.000465	.000160	.000532	0%
UR5e, 20	.001334	.002745	.001436	.003023	0%

Table 3: ABB retained-candidate replay diagnostics. Spearman is computed within snapshots with defined ranks; all intervals are hierarchical bootstrap 95% intervals.

Metric	Estimate	95% interval
Within-snapshot Spearman	0.976	[0.917, 1.000]
Pairwise concordance	0.988	[0.957, 1.000]
Relative realized regret	0.192%	[0, 0.481]%
20-step branch q RMSE	0.889 mrad	[0.705, 1.121] mrad

Model validation and CEM ranking. Tables 2–3 provide the principal held-out and replay diagnostics. Complete window, rollout, and excitation summaries are in `abb/model_validation/aggregate.csv`, `abb/model_validation/analysis_manifest.json`, and the corresponding `ur5e/model_validation` files; historical padded-history diagnostics are also retained under `abb/model_validation`. Counterfactual replay is in `abb/model_validation/formal_mpc_replay_aggregate.csv` and `claim_evidence/summaries/candidate_ranking.json`. It covers retained, projection-consistent online candidates only; its lower branch error must not be compared directly with the broader random-excitation validation.

Robustness and timing. ABB robustness applies L3/L6 single-factor payload, actuator-gain, TCP-force, and observation-noise perturbations. Per-condition/bootstrap results, timing, packet behavior, and projection activity are in `robustness_and_timing` and `abb/summaries`. Primary experiments had no planner failure or imposed joint, command-velocity, or command-acceleration violation; this does not imply a physical actuator limit.

Mechanism isolation. The frozen 20-case ABB matrix gives +2.82 mm for removing future alignment (95% CI [1.27, 4.39] mm), +654.60 mm for launch-anchor absolute replay ([528.68, 778.68] mm), and −0.95 mm for removing fast feedback ([−2.25, 0.28] mm), all relative to FullVirtual. A focused recurrent-history isolation changed TCP RMSE by −0.67 mm with an interval crossing zero. Thus the evidence supports temporal input consistency but does not establish an independent nominal tracking gain from recurrent-history advancement. Detailed projection/task-score and post-freeze effort diagnostics are released in the corresponding `abb/summaries` and `claim_evidence/summaries` files; they did not replace the frozen primary configuration.